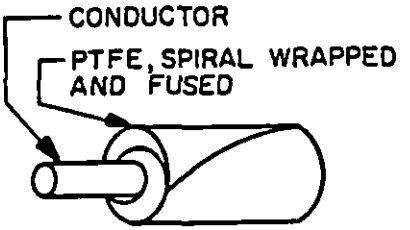
MIL-W-16878/26(NAVY)
10 August 1981
SUPERSEDING
MIL-W-16878/4A(NAVY) (IN PART)
5 July 1961
(See supersession data herein)

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE (PTFE) INSULATED, 260°C, 600 VOLTS, WRAPPED INSULATION

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the wire described herein shall consist of this document and the latest issue of MIL-W-16878:



SH 11884

FIGURE 1. Construction.

TABLE I. Construction details.

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/ 3/	Coating		Diameter (inch)	
					Min	Max
32	1 X 32	Copper	Nickel	0.0089	0.025	0.033
32	1 X 32	H.S.C.A.	Nickel	.0089	.025	.033
32	1 X 32	C.C. Steel	Nickel	.0089	.025	.033

See footnotes at end of table.

FSC 6145

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TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
					Diameter (inch)	
		Material 2/ 3/	Coating		Min	Max
32	7 X 40	Copper	Nickel	0.010	0.026	0.034
32	7 X 40	H.S.C.A.	Nickel	.010	.026	.034
30	1 X 30	Copper	Nickel	.0100	.026	.034
30	1 X 30	H.S.C.A.	Nickel	.0100	.026	.034
30	1 X 30	C.C. Steel	Nickel	.0100	.026	.034
30	7 X 38	Copper	Nickel	.012	.028	.036
30	7 X 38	H.S.C.A.	Nickel	.012	.028	.036
28	1 X 28	Copper	Nickel	.0126	.029	.037
28	1 X 28	H.S.C.A.	Nickel	.0126	.029	.037
28	1 X 28	C.C. Steel	Nickel	.0126	.029	.037
28	7 X 36	Copper	Nickel	.015	.031	.039
28	7 X 36	H.S.C.A.	Nickel	.015	.031	.039
26	1 X 26	Copper	Nickel	.0159	.032	.040
26	1 X 26	H.S.C.A.	Nickel	.0159	.032	.040
26	1 X 26	C.C. Steel	Nickel	.0159	.032	.040
26	7 X 34	Copper	Nickel	.019	.035	.043
26	7 X 34	H.S.C.A.	Nickel	.019	.035	.043
26	19 X 38	Copper	Nickel	.020	.035	.043
26	19 X 38	H.S.C.A.	Nickel	.020	.035	.043
24	1 X 24	Copper	Nickel	.0201	.036	.044
24	1 X 24	H.S.C.A.	Nickel	.0201	.036	.044
1/ 24	1 X 24	C.C. Steel	Nickel	.0201	.036	.044
24	7 X 32	Copper	Nickel	.024	.040	.048
24	7 X 32	H.S.C.A.	Nickel	.024	.040	.048
24	19 X 36	Copper	Nickel	.025	.040	.048
24	19 X 36	H.S.C.A.	Nickel	.025	.040	.048
22	1 X 22	Copper	Nickel	.0254	.041	.050
22	1 X 22	H.S.C.A	Nickel	.0254	.041	.050
1/ 22	1 X 22	C.C. Steel	Nickel	.0254	.041	.050
22	7 X 30	Copper	Nickel	.030	.046	.054
22	7 X 30	H.S.C.A.	Nickel	.030	.046	.054
22	19 X 34	Copper	Nickel	.032	.046	.054
22	19 X 34	H.S.C.A.	Nickel	.032	.046	.054
20	1 X 20	Copper	Nickel	.0320	.048	.056
20	1 X 20	H.S.C.A.	Nickel	.0320	.048	.056
20	7 X 28	Copper	Nickel	.038	.054	.062
20	7 X 28	H.S.C.A.	Nickel	.038	.054	.062
20	19 X 32	Copper	Nickel	.040	.054	.062
20	19 X 32	H.S.C.A.	Nickel	.040	.054	.062

See footnotes at end of table.

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TABLE I. Construction details. - Continued

Wire size	Stranding	Conductor		Conductor diameter (nominal) (inch)	Finished wire	
		Material 2/3/	Coating		Diameter (inch)	
					Min	Max
18	1 X 18	Copper	Nickel	0.0403	0.056	0.066
18	7 X 26	Copper	Nickel	.048	.064	.074
18	19 X 30	Copper	Nickel	.050	.064	.074
16	1 X 16	Copper	Nickel	.0508	.067	.081
16	19 X 29	Copper	Nickel	.057	.073	.087
14	19 X 27	Copper	Nickel	.072	.088	.102
12	19 X 25	Copper	Nickel	.091	.107	.121
12	37 X 28	Copper	Nickel	.089	.105	.119
10	37 X 26	Copper	Nickel	.111	.127	.141

1/ Inactive for new design.
2/ H.S.C.A. stands for high strength copper alloy.
3/ C.C. Steel stands for copper clad steel.

REQUIREMENTS:

Visual and mechanical inspection: Applicable.
Spark test: 3.4 kilovolts (kV).
Impulse dielectric test: 6.5 kV.
Tank test: 2.0 kV.
Dielectric withstanding voltage: 2.0 kV.
Insulation resistance: $IR = K \log_{10} D/d$

Where: IR = Minimum insulation resistance in megohms per 1000 feet at 20°C
K = 50,000
D = Maximum average diameter of finished wire
d = Conductor diameter

Cold bend: Condition 4 hours at minus 65°C ± 1°C (see table II).

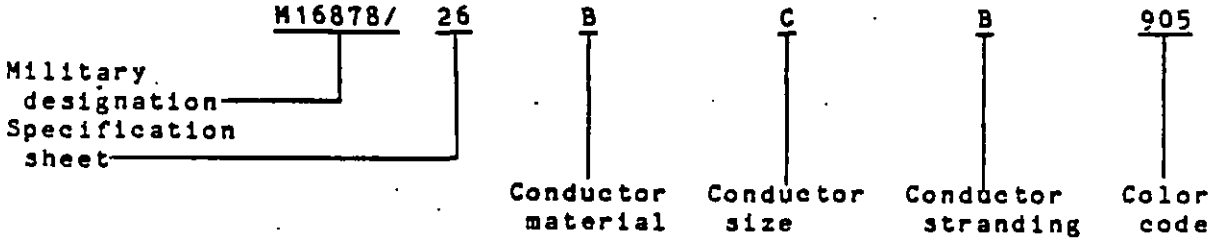
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TABLE II. Cold bend mandrel sizes.

Wire size	Cold bend mandrel (maximum diameter) (inches)
32 through 16	1
14 through 10	2

Concentricity: 70 percent (minimum).
Surface resistance: 5 megohms (minimum).
Wrap back: Applicable.
Heat resistance: Condition at 290°C.
Flammability: Not applicable.
Soldering: Applicable.
Heat aging: Not applicable.
Insulation tensile strength: 3000 pounds force per square inch (minimum).
Insulation elongation: 150 percent (minimum).
Stripe durability: Applicable.
Fungus resistance: Applicable.
Maximum direct current resistance of finished wire: See table I of MIL-W-16878.

Military part number (see MIL-W-16878):



Supersession data: The wire covered by this specification sheet supersedes MIL-W-16878/4A(NAVY), type E, 260°C rated, with tape wrapped insulation.

Review activities:
Navy - AS, EC
User activities:
Navy - OS, MC

Preparing activity:
Navy - SH
(Project 6145-N300-25)

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